TOOL MANUAL

ACZ 9185 MULTI-PURPOSE CUT-OFF MACHINE

---

1. DESCRIPTION

The **ACZ 9185 Cut-Off Machine** is a compact, bench-mounted electric cutting machine designed for performing precise and efficient cutting operations in workshop and laboratory environments. It is a versatile tool capable of cutting both metallic and non-metallic materials by using appropriate cutting blades.

This machine is widely used for fabrication work, material preparation, and general workshop applications due to its portability, ease of operation, and multi-purpose functionality.

IDENTIFICATION DETAILS

- **Tool Name:** Cut-Off Machine / Multi-Purpose Cutting Machine
- **Model Number:** ACZ-9185
- **Make / Brand:** ACZ (ACT Corporation Malaysia)
- **Machine Type:** Bench-mounted electric cutting machine
- **Blade Size:** 185 mm
- **Power Supply:** Single phase AC supply
- **Operation Type:** Manual downward cutting
- **Cooling System:** Air cooled motor

MAIN COMPONENTS

The machine consists of the following major components:

- Electric motor
- Cutting wheel / blade(Abrasive & toothed wood cutting blade)
- Protective blade guard
- Operating handle with switch
- Machine base
- Work holding vice / clamp
- Power cable

Each component plays an important role in ensuring safe and efficient operation.

2. APPLICATIONS

The ACZ 9185 Cut-Off Machine is used for cutting a variety of materials depending on the blade used.

Metal Cutting Applications

When fitted with an abrasive cutting wheel, the machine is used for cutting:

- Mild steel rods
- Metal pipes
- Flat bars
- Metal strips
- Small structural sections

Wood Cutting Applications

When fitted with a toothed wood cutting blade, the machine is used for cutting:

- Wooden strips
- Plywood sheets
- Wooden blocks
- Soft boards

This makes the machine suitable for multi-material cutting operations.

3. OPERATING PROCEDURE

The machine should be operated carefully by following the standard procedure.

Pre-Operation

The machine should be inspected before use. The blade must be checked for cracks, wear, or looseness. The guard and power cable must be in proper condition.

Blade Selection

The appropriate blade should be selected based on the material:

- Abrasive wheel for metal
- Toothed blade for wood

Workpiece Setup

The workpiece should be marked and securely clamped using the vice to prevent movement during cutting.

Machine Operation

The machine should be switched ON and allowed to reach full speed. The blade should then be lowered slowly onto the workpiece with steady pressure.

Post-Operation

After cutting, the machine should be switched OFF and the blade should be allowed to stop completely before removing the workpiece.

---

4. DO’s AND DON’Ts

Do’s

- Always wear safety goggles, gloves, and safety shoes.
- Inspect the machine before operation.
- Use the correct blade for the material.
- Clamp the workpiece firmly.
- Maintain proper posture while operating.
- Allow the blade to reach full speed before cutting.
- Switch OFF the machine after use.

Don’ts

- Do not use damaged or cracked blades.
- Do not operate without the guard.
- Do not hold the workpiece by hand.
- Do not use the wrong blade for the material.
- Do not apply excessive force.
- Do not wear loose clothing near the machine.
- Do not touch the blade immediately after use.

---

5. SAFETY MEASURES

Personal Safety : The operator must wear proper protective equipment such as safety goggles, gloves, and safety shoes. Loose clothing and accessories should be avoided.

Machine Safety : The blade guard must always be in position. The machine should be stable and all components should be properly tightened.

Operational Safety : Hands must be kept away from the rotating blade. The workpiece must be securely clamped. The blade should not be forced into the material.

Electrical Safety : The machine should be operated with a proper power supply. Damaged cables should not be used. The machine should not be operated with wet hands.

Fire Safety : Metal cutting produces sparks, so flammable materials must be kept away. Wood cutting produces dust and chips, so proper cleaning and ventilation are necessary.

---

6. IMAGE OF THE TOOL